

SAUMYA JAIN

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Summary

Data Science student with hands-on experience building production-grade ML, NLP, and LLM systems. Skilled in end-to-end RAG pipelines, semantic search, and grounded AI systems using Python, SBERT, ChromaDB, LangChain, and FastAPI. Strong foundation in supervised learning, recommendation systems, and anti-hallucination design.

Skills

Languages: Python, SQL
ML/DS: Scikit-learn, XGBoost, Random Forest, Pandas, NumPy, Feature Engineering
NLP/LLM: SBERT, NER, Embeddings, Semantic Search, RAG, LangChain, Hugging Face Transformers, Groq (Llama 3.1), Prompt Engineering
Tools & Backend: FastAPI, Git, Streamlit, Jupyter, VS Code
Databases: PostgreSQL, MongoDB, ChromaDB
Visualization: Matplotlib, Seaborn

Projects

- papertrail — PDF-Constrained Conversational Agent** | *Python, RAG, ChromaDB, SBERT, Groq* | 🐙
- Built an end-to-end RAG pipeline with PyMuPDF, LangChain chunking, SBERT embeddings, and ChromaDB — delivering strictly grounded responses with page citations and zero hallucination outside document scope.
 - Implemented a 0.25 cosine similarity refusal threshold rejecting out-of-scope queries — validated across 6 query types including multilingual (Spanish, Hinglish).
- Semantic Resume Screening System** | *Python, SBERT, NLP, FastAPI* | 🐙
- Built end-to-end screening pipeline using **SBERT embeddings** and a custom **NER extractor**, achieving **85%+ matching accuracy** and cutting screening time to **under 2 seconds**.
 - Developed a **FastAPI** backend for PDF ingestion, ranked candidate scoring, and explainable output — deployable as a standalone REST service.
- Customer Churn Prediction System** | *Python, XGBoost, Scikit-learn, SQL* | 🐙
- Engineered features using advanced SQL (CTEs, window functions) on a **normalized PostgreSQL schema** simulating a production banking environment.
 - Compared **XGBoost, Random Forest, and Logistic Regression**; deployed best AUC-ROC model via an automated **ETL + batch inference** pipeline.
- Time-Aware Movie Recommendation System** | *Python, Collaborative Filtering* | 🐙
- Engineered a **temporal collaborative filtering** model weighting recent interactions more heavily, outperforming standard matrix-factorization baselines.
 - Benchmarked SVD, KNN, and time-decay CF using **NDCG and Precision@K** across a modular train/test pipeline split by timestamp.
- Smart Attendance Dashboard — AcademiQ** | *FastAPI, React, Scikit-learn, PostgreSQL* | 🐙
- Built a **full-stack AI platform** (React + FastAPI) for real-time attendance tracking with ML models flagging **at-risk students** at **80%+ recall**.
 - Automated CSV ingestion into **PostgreSQL**, eliminating manual data entry and enabling scalable batch updates for institutional use.

Education

Scaler School of Technology <i>Bachelor's + Masters in Computer Science</i>	2024–2028
BITS Pilani <i>Bachelor's in Computer Science</i>	2024–2027

Achievements

- Ranked top 10 out of 400 students for a Machine Learning project — evaluated on accuracy, feature engineering, and system performance.

Achievements & Certifications

- Machine Learning by Andrew Ng** – Coursera
- IBM Data Science Professional Certificate** – Coursera